Unit 10, Lesson 4: Is It a Function?

Benchmark

MA.8.F.1.1 Given a set of ordered pairs, a table, a graph, or mapping diagram, determine whether the relationship is a function. Identify the domain and range of the relation.

Learning Target

☐ I can determine whether a relation is a function.

10.4.1 Warm-Up: Sticking With It

Engineering students learn many lessons both in and out of the classroom. One student shared what she learned: "Sometimes there is a lot of pressure that can cause unwanted strain, but it is what happens at the point of failure that reveals the amount of strength that is really inside."

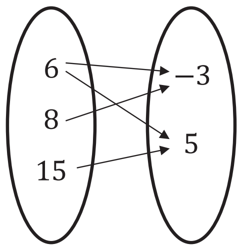
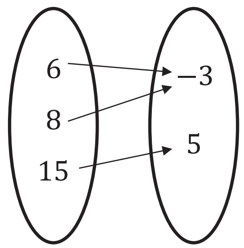
1. What is one life lesson you have learned this school year?

2. When is a time when you have had to stick with something that was hard in order to reach a goal?

10.4.2 Guided Practice: Is It a Function?

A function is a mathematical relation for which each element in the domain corresponds to exactly one element in the range.

1. Two relations are shown below.

Relation A	Relation B
	

- Mila said that neither Relation A nor Relation B is a function.
- K’Drien said that Relation A is not a function, but reasoned that Relation B is a function.

Complete the statements.

I agree with ○ Mila. ○ K’Drien.

I know that Relation B ○ is ○ is not a function because

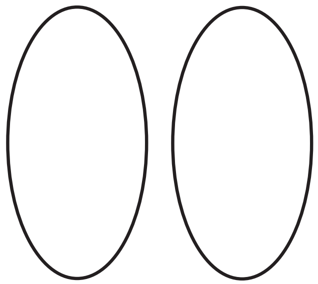
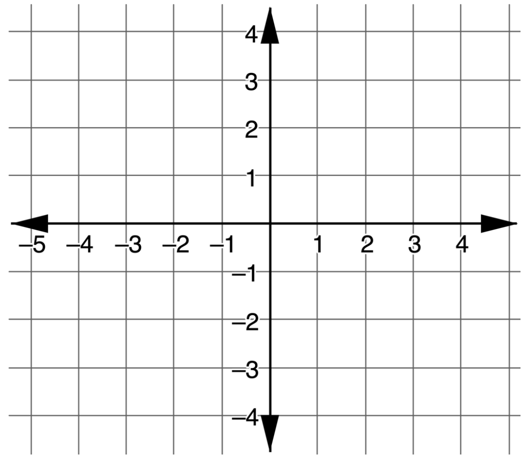
each element in the ○ domain ○ range ○ does ○ does not

correspond with exactly one element in the ○ domain. ○ range.

2. A relation is given as a set of ordered pairs.

$$\{(2,4),(1,1),(-1,4),(-3,1),(0,-3)\}$$

a. Determine with your partner who will be Partner A and who will be Partner B. Represent the relation using the method associated with your letter.

Mapping Diagram (Partner A)	Graph (Partner B)
	

b. Check your partner’s representation. Once you both agree that each representation is correct, complete the other representation in your book.

c. Complete the statements to explain how each representation shows whether the relation is a function.

The **mapping diagram** has

- ○ only one arrow to
- ○ only one arrow from
- ○ more than one arrow to
- ○ more than one arrow from

each element in the

- ○ domain
- ○ range

to its corresponding element in the

- ○ domain,
- ○ range,

therefore the relation

- ○ is
- ○ is not

a function.

The **graph** demonstrates the relation

- ○ is
- ○ is not

a function because

- ○ each x-value
- ○ at least one x-value

corresponds to

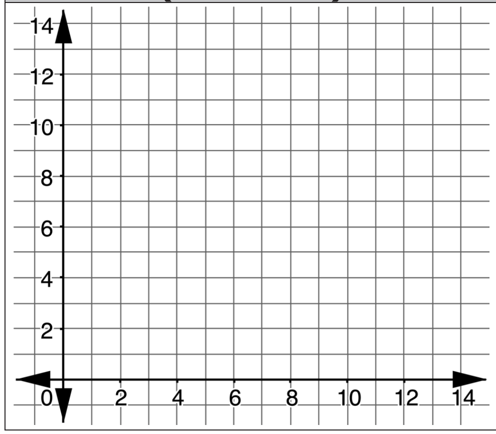
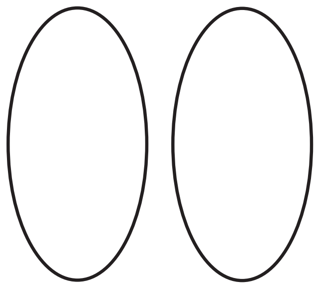
- ○ exactly one y-value
- ○ more than one y-value

on the graph.

3. The table below gives the number of letters of the winning word of the Scripps National Spelling Bee, and the number of people killed by venomous spiders for the years 1996-2006 (CDC).

	1999	**2000**	**2001**	**2002**	**2003**	**2004**	**2005**	**2006**
Letters in winning word	9	8	11	12	11	13	12	9
People killed by venomous spider	6	5	5	10	8	14	10	4

a. Using the same partner letters as in the previous problem, represent the relation with a mapping diagram and as a graph.

Graph (Partner A)	**Mapping Diagram (Partner B)**
	

b. Check your partner's representation. Once you both agree that each representation is correct, then complete the other representation.

c. Use the definition of a function to justify whether the relationship between the number of letters of the winning word of the Scripps National Spelling Bee and the number of people killed by venomous spiders is a function or not.

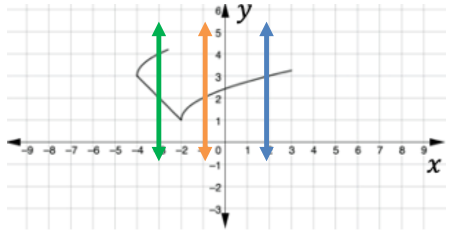
10.4.3 Exploration: Vertical Line Test

The vertical line test can be used to determine if a relation represented by a graph is a function.

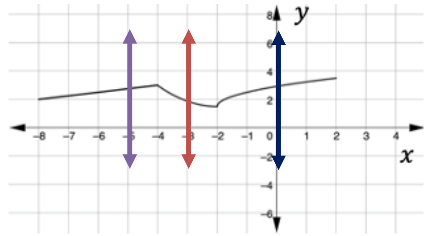
The vertical lines drawn on the graph of a function show whether each x-value corresponds to exactly one y-value.

A graph of a relation and a function are shown with three different vertical lines drawn on each.

Relation



Function

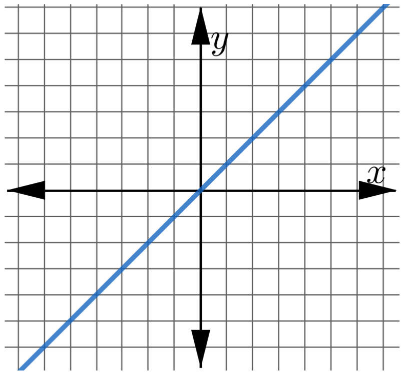
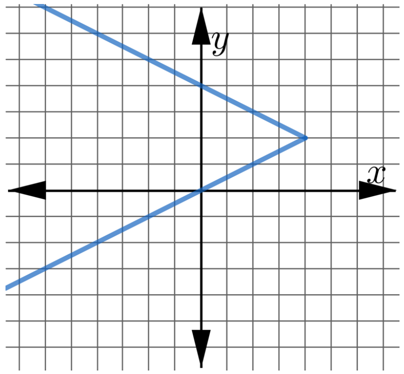
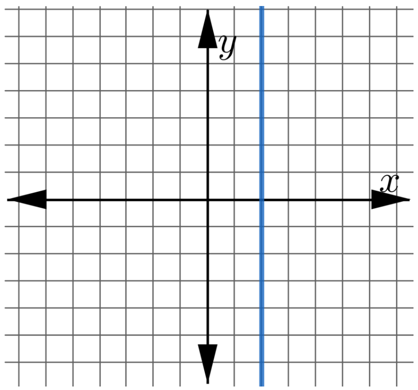
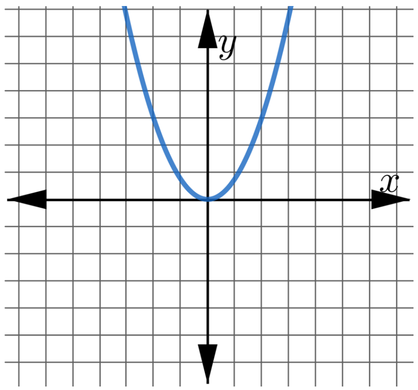


1. Complete the explanation that describes how the vertical line test can be used to show the relation represented by a graph is not a function.

The [○ blue ○ orange ○ green] line shows that the relation is not a function because the line demonstrates that for the x-value of [○ $^{-}3$ ○ $^{-}1$ ○ 2] there is more than one y-value.

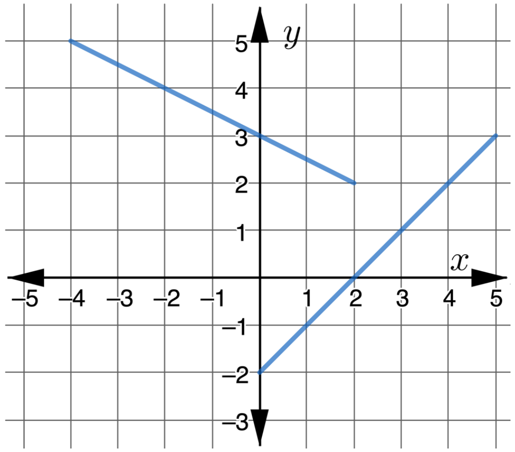
2. Sort the relations into each category by writing the letters that label the relations in the table.

Functions	Not Functions

A	B
	
C	**D**
	
E	**F**
$\{(2,7),(3,7),(2,-1),(7,3)\}$	$\{(-5,-1),(9,3),(4,-1),(-2,6)\}$
G	**H**
$\{(1,9),(-2,8),(0,12),(-6,0)\}$	$\{(-5,8),(6,-4),(2,11),(-5,-7)\}$

10.4.4 Your Turn!

1. A group of students were discussing whether the relationship on the graph is a function.



***Ed*:** The relationship is not a function because if I draw a horizontal line at $y = 2$, the horizonal line crosses the graph at two points.

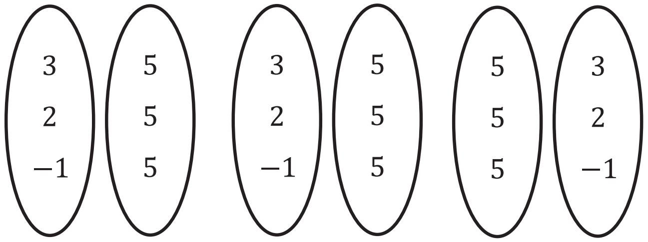
***Debra*:** The relationship is a function because it is a graph of two different lines.

***Alejandro*:** The relationship is not a function because $x = 1$ has two different y-values.

Which students are **not** correct and what is the flaw in their argument?

2. Consider the following incomplete mapping diagrams.

Diagram A Diagram B Diagram C



a. Complete Diagram A so that it is a function.

b. Is it possible to complete Diagram B so that it is not a function? If so, complete the diagram to show a relation, but not a function. If not, justify your reasoning.

c. Is it possible to complete the mapping diagram for Diagram C so it represents a function? If so, complete the diagram to show a function. If not, justify your reasoning.

3. Select all the relations that are **not** functions.

- ☐ $\{(1,3),(3,7),(5,11),(7,15),(9,19)\}$
- ☐ $\{(1,3),(1,7),(5,11),(5,15),(9,19)\}$
- ☐ $\{(-2,4),(-1,1),(0,0),(1,1),(2,4)\}$
- ☐ $\{(2,4),(1,1),(0,0),(1,-1),(2,-4)\}$
- ☐ $\{(6,3),(4,1),(2,1),(0,-1),(-2,-3)\}$
- ☐ $\{(1,3),(3,7),(3,11),(7,15),(9,19)\}$
- ☐ $\{(1,3),(3,7),(5,11),(9,15),(9,19)\}$

10.4.5 Wrap-Up: Error Analysis

The table shows the gold-medal winners of the 100-meter dash in the Olympic games since 1980.

Winner	Height (meters)	Time (seconds)
Allan Wells (GBR)	1.83	10.25
Carl Lewis (USA)	1.88	9.99
Carl Lewis (USA)	1.88	9.92
Linford Christie (GBR)	1.87	9.96
Donovan Bailey (CAN)	1.85	9.84
Maurice Greene (USA)	1.76	9.87
Justin Gatlin (USA)	1.86	9.85
Usain Bolt (JAM)	1.95	9.69
Usain Bolt (JAM)	1.95	9.63
Usain Bolt (JAM)	1.95	9.81

Both Olivia and Liam wrote an explanation about whether the table of values represents a function.

Olivia's Explanation	Liam's Explanation
The table of values of the gold-medal winners is a function because the winners competed in different years.	The table of values of the gold-medal winners is a function. Although there are some inputs that give different outputs, those inputs are for the same person.

1. Complete the statements to evaluate both explanations.

I agree with
- ○ Liam.
- ○ Olivia.
- ○ neither Liam nor Olivia.

- ○ Liam did not use
- ○ Olivia did not use
- ○ Neither Liam nor Olivia used

the correct
- ○ domain
- ○ range

to determine if the relation in the table is a function.